WORKED SOLUTION

STEP I 2014, Question 6

Pure • Logistic Map Recurrences

STEP I 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 8.7, *Chaos You Can Solve*: https://beyondstep.keithmcnulty.org/ch08/solvable-chaos.html

The key idea

The expression $4x(1-x)$ looks awkward until you put $x = \sin^2\theta$: then $1 - x = \cos^2\theta$ and $4\sin^2\theta\cos^2\theta = (2\sin\theta\cos\theta)^2$ is a double-angle formula in disguise. Each step of the recurrence simply doubles the angle. Part (ii) is then about recognising that *any* quadratic recurrence can be turned into this one by a linear change of variable, as long as one condition on the coefficients holds.

Solution

Part (i)

With $u_0 = \sin^2\theta$ we have $1 - u_0 = \cos^2\theta$, so

$$u_1 = 4\sin^2\theta\cos^2\theta = (2\sin\theta\cos\theta)^2 = \sin^2 2\theta.$$

This has exactly the same form as u_0, with θ replaced by 2θ, so the same calculation gives

$$u_2 = 4\sin^2 2\theta\ \cos^2 2\theta = \sin^2 4\theta.$$

The angle doubles at each step, so we conjecture

$$u_n = \sin^2(2^n\theta) \qquad (n \geq 0).$$

Proof by induction. For $n = 0$ this reads $u_0 = \sin^2\theta$, which is true. Suppose $u_k = \sin^2(2^k\theta)$

for some $k \geq 0$. Then $1 - u_k = \cos^2(2^k\theta)$, and

$$\begin{aligned} u_{k+1} &= 4\sin^2(2^k\theta)\cos^2(2^k\theta) \\ &= \big(2\sin(2^k\theta)\cos(2^k\theta)\big)^2 \\ &= \sin^2(2^{k+1}\theta), \end{aligned}$$

using $\sin 2A = 2\sin A\cos A$ with $A = 2^k\theta$. So the result for k implies the result for $k+1$, and by induction $u_n = \sin^2(2^n\theta)$ for all $n \geq 0$.

Part (ii)

Put $v_n = \alpha u_n + \beta$ with $\alpha \neq 0$ (we need $\alpha \neq 0$ so that we can recover u_n from v_n). Substituting into the recurrence for v_n:

$$\alpha u_{n+1} + \beta = -p(\alpha u_n + \beta)^2 + q(\alpha u_n + \beta) + r.$$

Expanding and dividing by α,

$$u_{n+1} = -p\alpha\, u_n^2 + (q - 2p\beta)\, u_n + \frac{-p\beta^2 + (q-1)\beta + r}{\alpha}.$$

We want this to be $u_{n+1} = 4u_n - 4u_n^2$, so we compare coefficients.

- u_n^2: $p\alpha = 4$, so $\alpha = \dfrac{4}{p}$ (fine, since $p \neq 0$, and then $\alpha \neq 0$ too).
- u_n: $q - 2p\beta = 4$, so $\beta = \dfrac{q-4}{2p}$.
- constant: $-p\beta^2 + (q-1)\beta + r = 0$.

The first two fix α and β; the third is then a condition on p, q, r. Substituting $\beta = (q-4)/(2p)$ and multiplying through by $4p$:

$$\begin{aligned} & 0 = -p \cdot \frac{(q-4)^2}{4p^2} + (q-1)\frac{q-4}{2p} + r \\ \iff \quad & 0 = -(q-4)^2 + 2(q-1)(q-4) + 4pr \\ & \quad = (q-4)\big(-(q-4) + 2(q-1)\big) + 4pr \\ & \quad = (q-4)(q+2) + 4pr \\ & \quad = q^2 - 2q - 8 + 4pr. \end{aligned}$$

So if $4pr = 8 + 2q - q^2$, then with $\alpha = 4/p$ and $\beta = (q-4)/(2p)$ all three coefficients match, and the sequence $u_n = (v_n - \beta)/\alpha$ satisfies $(*)$, as required.

The particular sequence. Here $p = 1$, $q = 2$, $r = 2$. The condition holds: $4pr = 8$ and $8 + 2q - q^2 = 8 + 4 - 4 = 8$. We get

$$\alpha = 4, \qquad \beta = \frac{2-4}{2} = -1, \qquad \text{so} \qquad v_n = 4u_n - 1.$$

Ask yourself Part (i) needs $u_0 = \sin^2\theta$. Can we find a suitable θ here?

Since $v_0 = 1$, we need $u_0 = (1+1)/4 = \frac{1}{2}$, and $\frac{1}{2} = \sin^2\frac{\pi}{4}$. So $\theta = \pi/4$ works, and part (i) gives $u_n = \sin^2(2^n\pi/4) = \sin^2(2^{n-2}\pi)$. Therefore

$$v_n = 4\sin^2\left(2^{n-2}\pi\right) - 1 = 1 - 2\cos\left(2^{n-1}\pi\right),$$

using $2\sin^2 A = 1 - \cos 2A$. Evaluating:

- $n = 0$: $v_0 = 4\sin^2\frac{\pi}{4} - 1 = 1$;
- $n = 1$: $v_1 = 4\sin^2\frac{\pi}{2} - 1 = 3$;
- $n \geq 2$: 2^{n-2} is an integer, so $\sin(2^{n-2}\pi) = 0$ and $v_n = -1$.

So the sequence is

$$1,\ 3,\ -1,\ -1,\ -1,\ \ldots$$

that is, $v_0 = 1$, $v_1 = 3$ and $v_n = -1$ for all $n \geq 2$.

Checks

Iterate the recurrence directly: $v_1 = -1 + 2 + 2 = 3$, $v_2 = -9 + 6 + 2 = -1$, and $v_3 = -1 - 2 + 2 = -1$; since -1 maps to itself, the sequence stays at -1, as the formula says. For part (i), try $\theta = \pi/6$: $u_0 = \frac{1}{4}$, $u_1 = 4 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{4} = \sin^2\frac{\pi}{3}$, and $u_2 = 4 \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{4} = \sin^2\frac{2\pi}{3}$, as predicted. The formula $u_n = \sin^2(2^n\theta)$ was also confirmed by computer for many random angles and the first several terms.

Common slips

- Writing $u_n = \sin^2(2n\theta)$: the angle doubles each time, so it grows like 2^n, not $2n$.
- Forgetting to check that the starting value $u_0 = (v_0 - \beta)/\alpha$ lies in $[0, 1]$, so that it really can be written as $\sin^2\theta$.
- Treating the constant-term equation as a third unknown to solve for, rather than as the condition the question asks you to reach.

Beyond STEP

The recurrence $u_{n+1} = 4u_n(1 - u_n)$ is the logistic map at its most extreme parameter, a standard example of chaos: since $u_n = \sin^2(2^n\theta)$, writing θ/π in binary shows that each step just shifts the binary digits one place, so two nearby starting values separate exponentially fast. The substitution in part (ii) is a *conjugacy*, the idea that two dynamical systems are really the same after a change of coordinates. Strogatz's *Nonlinear Dynamics and Chaos* is a very readable introduction to these ideas.

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